Color design and typography

UI Hall of Fame or Shame?

Source: UI Hall of Shame

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color-blind users ?

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Color Blindness

- Red-green color blindness (protanopia & deuteranopia)
  - 8% of males
  - 0.4% of females
- Blue-yellow color blindness (tritanopia)
  - Far more rare (~50 people in a million)
- Guideline: don’t depend solely on color distinctions
  - use redundant signals: brightness, location, shape

spelling errors vs grammar errors

gwegeg. This are a test.

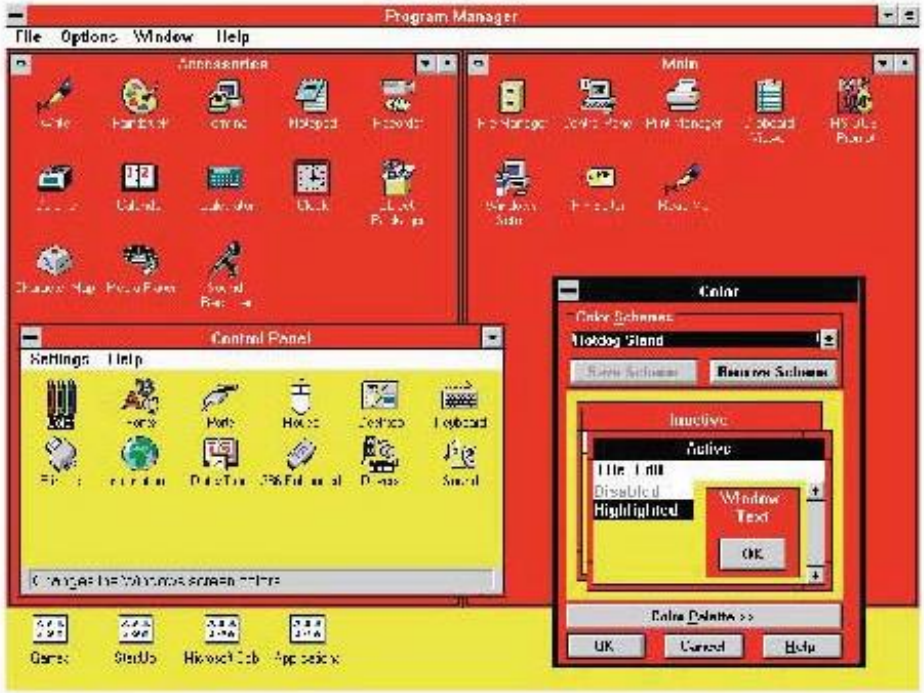
Chromatic Aberration

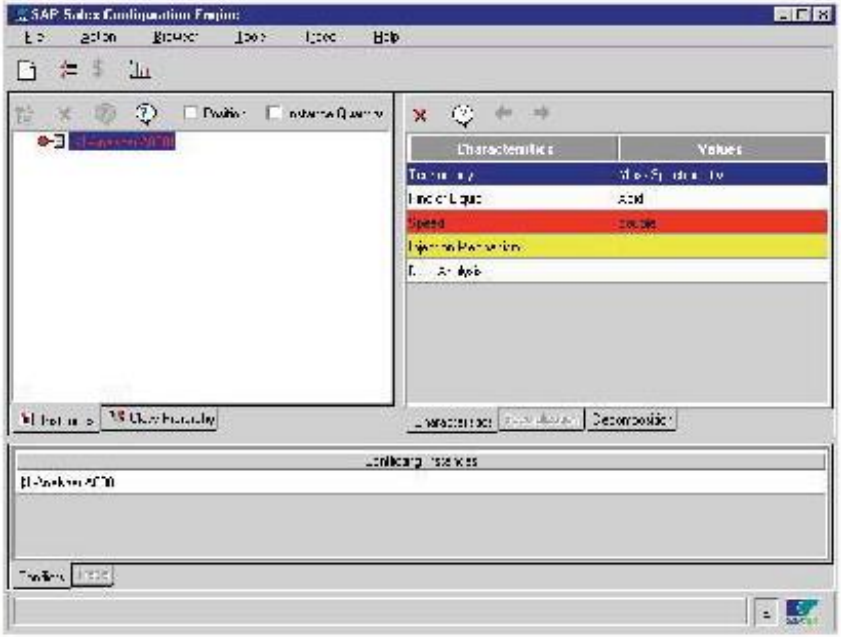
- Different wavelengths focus differently
  - Highly separated wavelengths (red & blue) can't be focused simultaneously
- Guideline: don't use red-on-blue text
  - It looks fuzzy and hurts to read

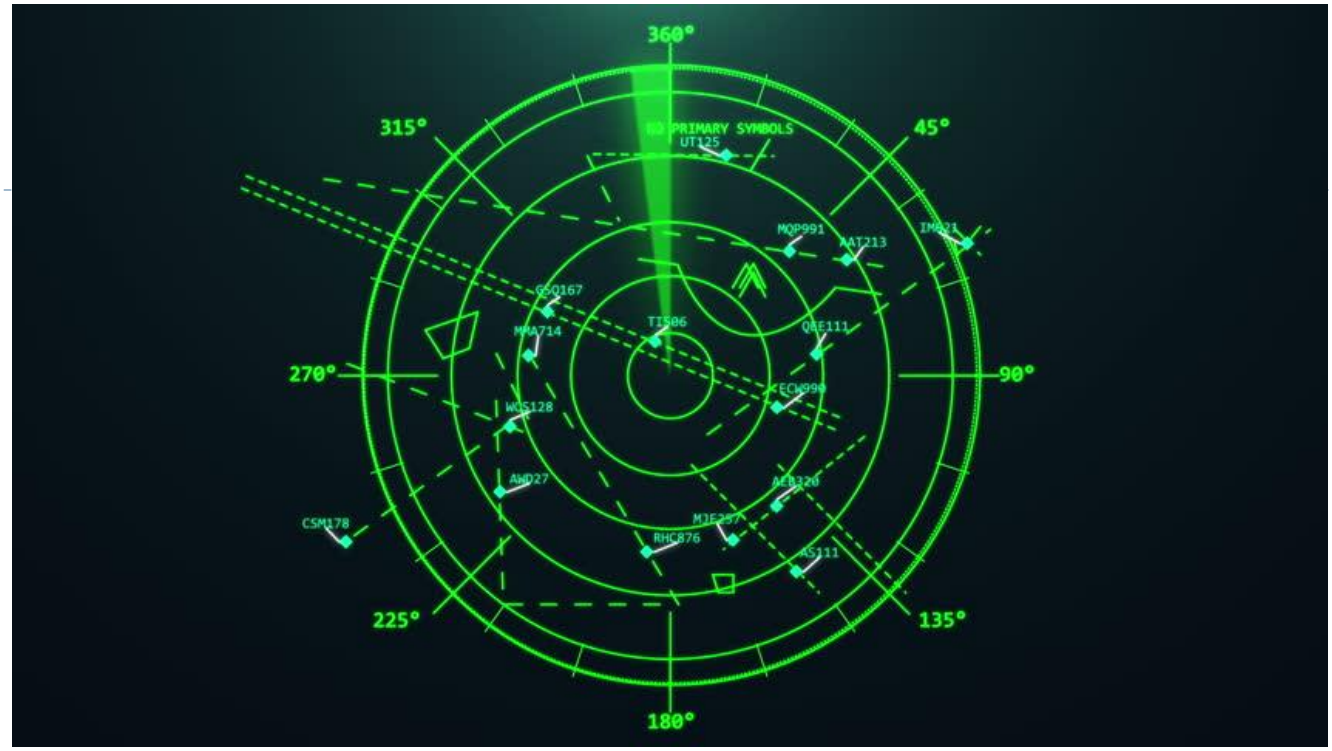
Color Guidelines

- Avoid saturated colors
- Use few colors
- Be consistent with expectations

Avoid Saturated Colors

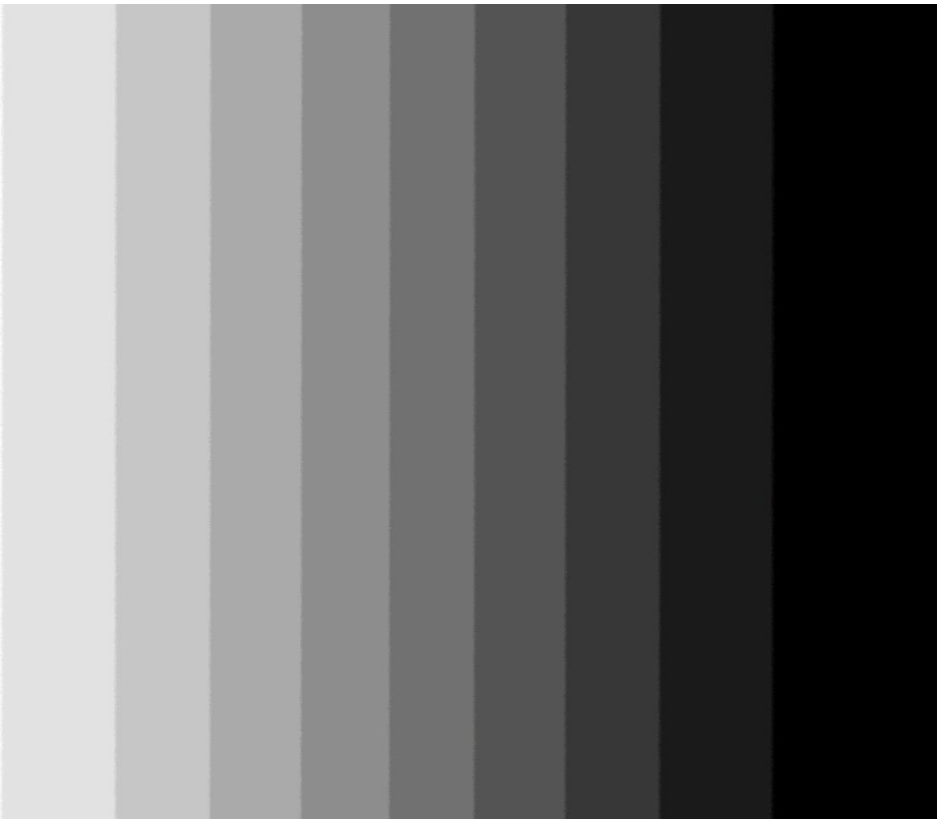






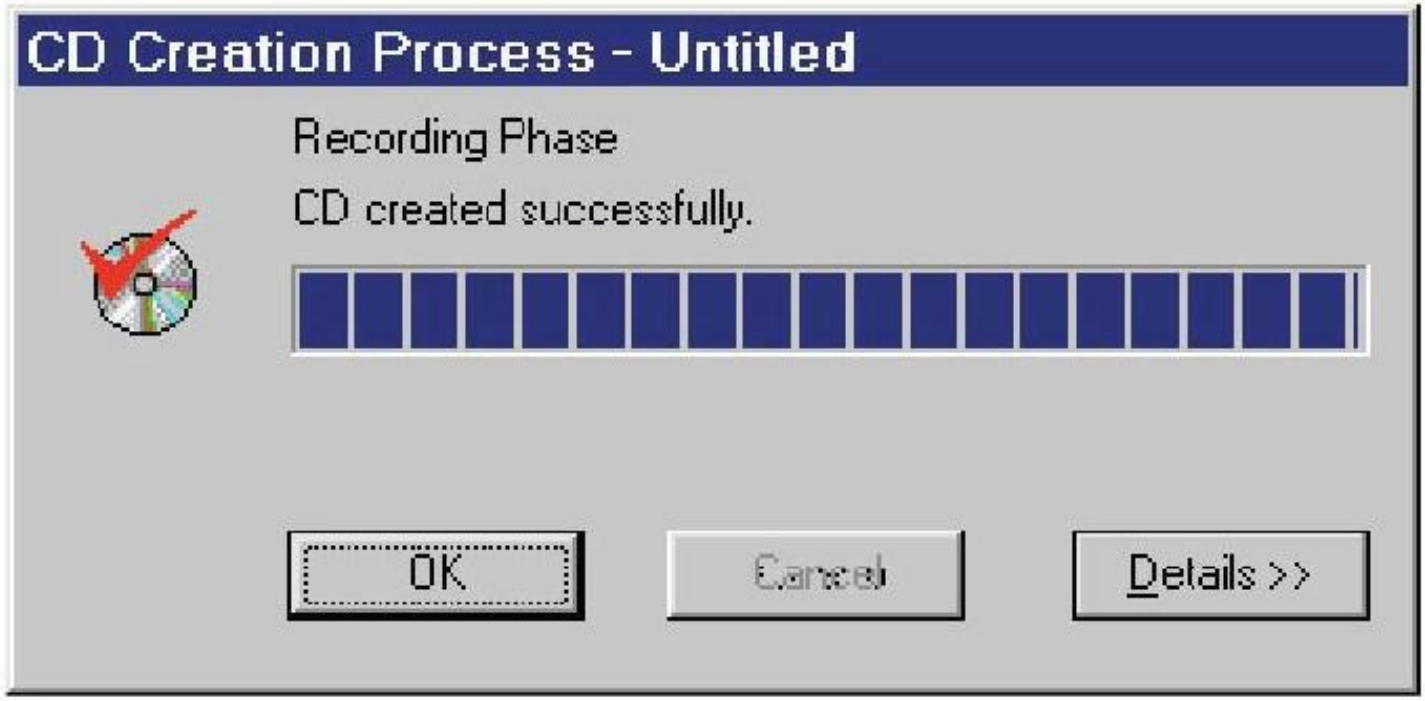
One study found that air traffic controllers who viewed strongly saturated green text on their ATC interfaces for many hours had trouble seeing pink or red (the other end of the red/green color channel) for up to 15 minutes after their shift was over.

Shades of gray have minimum saturation



Use Few Colors

Be Consistent With Expectations



Font Selection

- Simplicity & contrast
  - Don’t use more than 2 or 3 typefaces
    - E.g., one for body text, one for display text
  - Use size, weight, style (e.g. italic/small caps), hue to establish essential contrasts
    - But 4-5 font varieties should be enough

Summary

- Don’t rely solely on color distinctions
  - Color blindness is common
- Keep your color design simple
  - Use few colors, weakly saturated
- Whitespace matters for text
  - Use generous margins, line spacing, short lines
- Keep font choices simple
  - Few typefaces, few sizes and styles